

sin() Stays sin(); Angles Become Plural v3 Evil-Terse: One Relation, Paired $\mathbb{R}^7$, Different Bodies

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Abstract

Special relativity already contains the seed. For one speed state,

$$\beta = \frac{v}{c} = \sin \theta = \tanh \chi, \quad \gamma = \sec \theta = \cosh \chi,$$

where $\theta = \text{gd } \chi$. Nothing is wrong with `sin()`. The mistake is requiring every representation of the same geometric relation to wear the same kind of angle. General Peace Dynamics (GPD) takes the next step in real Euclidean $\mathbb{R}^7$: each spatial label $i \in \{x, y, z\}$ belongs to a conjugate pair (i_t, t_i) stitched by τ . The old one-axis (c_i, v_i, c) triangle is then read not as evidence for a mysterious component of light-speed, but as a projected, normalized image of a paired $\mathbb{R}^7$ direction. Circular vantage tilt, rapidity, reciprocal aperture, and gamma become different bodies of one relation. Same relation. Different body.

1. Start where the rotation actually lives

The arena is not merely “a seven-dimensional manifold.” The GPD coordinate choice is real Euclidean seven-space,

$$\boxed{\mathcal{M}^7 \equiv \mathbb{R}^7} \quad (1)$$

$$\begin{aligned} \mathbb{R}^7 &= (\tau, (x_t, t_x), (y_t, t_y), (z_t, t_z)) \\ &= (\tau, (i_t, t_i)_{i \in \{x, y, z\}}). \end{aligned}$$

The notation (x_i, t_i) remains a conventional phase-space-like reading; (i_t, t_i) exposes the architectural symmetry. The ordering is the point: $i_t \leftrightarrow t_i$ is a conjugate pairing, not decoration.

Why care about $\mathbb{R}^7$? GPD begins from real rotation. The familiar norm-compatible vector cross product is nontrivial in $\mathbb{R}^3$ and $\mathbb{R}^7$ (with the trivial $\mathbb{R}^0, \mathbb{R}^1$ cases completing the familiar 0, 1, 3, 7 sequence). Thus the recursive dimensional grammar

$$\mathbb{R}^0 \rightarrow \mathbb{R}^1 \rightarrow \mathbb{R}^3 \rightarrow \mathbb{R}^7$$

is not arbitrary bookkeeping. It is the native real-space ladder on which GPD’s rotational construction is built. At the coordinate level,

$$7 = 1 + 2(3) : \quad \tau + (x_t, t_x) + (y_t, t_y) + (z_t, t_z).$$

Call the resulting recursive construction biquadoctree if desired; the mathematical burden is to make the recursion precise without losing the normed rotational structure.

2. The one-axis SR seed

Take one ordinary collinear SR state,

$$\beta_v := \frac{v}{c}, \quad \gamma_v := \frac{1}{\sqrt{1 - \beta_v^2}}. \quad (2)$$

The original GPD SR sketch encoded the same state by a right triangle

$$v^2 + c_x^2 = c^2, \quad \beta_v^2 + \beta_c^2 = 1, \quad \beta_c := \frac{c_x}{c}. \quad (3)$$

It then used the overloaded vantage-tilt convention $\alpha \mid \delta$ — temporal $\mid$ spatial — and the reciprocal quantities

$$\alpha := \frac{1}{\beta_v}, \quad \delta := \frac{1}{\beta_c}. \quad (4)$$

The apparently strange dual gammas are already forced by Eq. (??):

$$\gamma_v = \frac{1}{\beta_c} = \delta, \quad \gamma_c = \frac{1}{\beta_v} = \alpha. \quad (5)$$

This is why gamma matters here. The reciprocal “tilt angles” and the two gamma-like factors were never separate curiosities; they are the same orthogonal complementarity read from opposite legs.

Now introduce the ordinary circular angle θ of the normalized triangle:

$$\beta_v = \sin \theta, \quad \beta_c = \cos \theta. \quad (6)$$

Then

$$\boxed{\alpha = \csc \theta, \quad \delta = \sec \theta, \quad \frac{1}{\alpha^2} + \frac{1}{\delta^2} = 1.} \quad (7)$$

The old line “ $\sin \alpha = 1/\alpha$ ” is therefore not new trigonometry. It is compressed notation that forgot the map from reciprocal aperture α back to circular angle θ . The missing map is the hole in the notation.

3. Gudermann: the bridge was already there

Let rapidity be

$$\chi := \text{artanh } \beta_v. \quad (8)$$

The Gudermannian gives the exact circular–hyperbolic bridge

$$\theta = \text{gd } \chi, \quad \sin \theta = \tanh \chi, \quad \cos \theta = \text{sech } \chi. \quad (9)$$

Hence one SR state admits the exact identity ledger

$$\boxed{\beta_v = \sin \theta = \tanh \chi = \alpha^{-1},} \quad (10)$$

$$\boxed{\beta_c = \cos \theta = \text{sech } \chi = \delta^{-1},} \quad (11)$$

$$\boxed{\alpha = \csc \theta = \coth \chi, \quad \delta = \sec \theta = \cosh \chi.} \quad (12)$$

Every equality above is ordinary mathematics. GPD’s move is ontological and architectural: promote the shared state to one vantage relation $a \in \mathcal{A}$, and treat $\theta, \chi, (\alpha, \delta)$, and β as different readable bodies of a . `sin()` stays ordinary; its argument has a type and a chart.

4. Give the conjugate analogy its chance

The symbol c_x is suspicious because c already names the invariant speed. Promoting c_x to (c_x, c_y, c_z) falsely invites the reading “Cartesian components of c .” Renaming the symbol does not solve the geometry. Restore the pair that may have been projected out.

For each i , take the paired plane inside Eq. (??),

$$\Pi_i := \text{span}\{\mathbf{e}_{i_t}, \mathbf{e}_{t_i}\} \subset \mathbb{R}^7. \quad (13)$$

After the physically correct dimensional normalization is fixed, write its unit tangent orientation as

$$\boxed{\mathbf{u}_i = \beta_{i_t} \mathbf{e}_{i_t} + \beta_{t_i} \mathbf{e}_{t_i}, \quad \|\mathbf{u}_i\| = 1.} \quad (14)$$

Thus

$$\boxed{\beta_{i_t}^2 + \beta_{t_i}^2 = 1.} \quad (15)$$

This is the canon spark. The one-axis triangle is no longer fundamentally “velocity plus complementary light-speed.” It is the normalized shadow of one conjugate $\mathbb{R}^7$ direction.

The physical map from coordinates to the two dimensionless tangent components is a derivational obligation, not a reason to abandon the pairing. A natural candidate uses τ and light-distance $c d\tau$ to normalize the two members; another may reveal that a literal differential quantity d_i is the right physical object. The notation must follow that derivation. It must not prejudice it.

5. $\alpha_i \mid \delta_i$ returns home

Choose orientation so that

$$\beta_{t_i} = \sin \theta_i, \quad \beta_{i_t} = \cos \theta_i. \quad (16)$$

Then define the reciprocal vantage apertures

$$\alpha_i := \frac{1}{\beta_{t_i}}, \quad \delta_i := \frac{1}{\beta_{i_t}}. \quad (17)$$

The original temporal|spatial $\alpha_i \mid \delta_i$ language is now attached to a real paired plane rather than to a mysterious c_i .

Define gamma-like factors for each normalized member:

$$\gamma_{i_t} := \frac{1}{\sqrt{1 - \beta_{t_i}^2}}, \quad \gamma_{t_i} := \frac{1}{\sqrt{1 - \beta_{i_t}^2}}. \quad (18)$$

Equation (??) immediately gives, on the positive branch,

$$\boxed{\alpha_i = \gamma_{i_t}, \quad \delta_i = \gamma_{t_i}.} \quad (19)$$

The crossed gamma relation is therefore structural: the gamma associated with one member is the reciprocal magnitude of the other. What looked like $\gamma_c \mid \gamma_v$ becomes conjugate geometry.

6. Euclidean rotation is the wager, not a footnote

The point of introducing θ_i was never merely to draw a cute triangle. In Π_i , Eq. (??) is an ordinary Euclidean unit direction, so a vantage change can be represented locally by an ordinary rotation

$$\mathbf{u}'_i = R_i(\theta_i)\mathbf{u}_i, \quad R_i(\theta_i) \in SO(2) \subset SO(7). \quad (20)$$

That is the original attraction: special-relativistic projection effects may be shadows of real rotations in a lifted real space, followed by the required projection/squeeze back to observer spacetime.

But the Gudermannian makes the test sharp. Collinear boosts compose by rapidity,

$$\chi_{12} = \chi_1 + \chi_2, \quad (21)$$

while $\theta = \text{gd } \chi$ does not obey ordinary angle addition:

$$\theta_{12} = \text{gd}(\text{gd}^{-1} \theta_1 + \text{gd}^{-1} \theta_2) \neq \theta_1 + \theta_2. \quad (22)$$

Therefore GPD's Euclidean-rotation program cannot end at "boost = rotate by θ ." The full lift-rotate-squeeze/projection architecture must carry the non-linear chart composition, or the additive rotation parameter must be a different body of the same vantage relation. This is not an embarrassment; it is the constraint that makes the proposal calculable.

7. Three paired slices are not three Lorentz gammas

For ordinary three-velocity,

$$\gamma(\mathbf{v}) = \frac{1}{\sqrt{1 - (v_x^2 + v_y^2 + v_z^2)/c^2}}. \quad (23)$$

The local paired quantities γ_{t_i} and γ_{i_t} are not three copies of Eq. (??). They belong to the three conjugate planes inside one $\mathbb{R}^7$ construction. The research problem is the reclosure: how the three locally circular paired directions, stitched by τ , project to the single conventional $3 + 1$ Lorentz state.

That is precisely why Eq. (??) must precede the componentwise notation. Without the paired $\mathbb{R}^7$ architecture, β_c looks invented. With it, the old complement becomes a clue that one half of the pair was projected away.

8. Plural angles, finally stated without apology

The weak statement is reparameterization. The stronger GPD statement is a construction principle:

$$\boxed{\text{one vantage relation} \longrightarrow \text{plural bodies}} \quad (24)$$

$$\begin{aligned} \chi_i &: \text{hyperbolic/additive,} \\ \theta_i &: \text{circular/rotational,} \\ (\alpha_i, \delta_i) &: \text{reciprocal/aperture,} \\ (\beta_{i_t}, \beta_{t_i}) &: \text{paired tangent.} \end{aligned}$$

The bodies obey different coordinate laws because they are not the same coordinate. They are required to close on the same relation.

This turns the original notational offense into the thesis:

$$\boxed{\sin() \text{ was fine. We passed it the wrong kind of angle.}}$$

And the paired $\mathbb{R}^7$ extension sharpens the second half:

$$\boxed{\text{Do not rename the missing complement. Restore its conjugate body.}}$$

9. The immediate derivational obligations

The next calculation must determine the dimensionally exact map $(i_t, t_i, \tau) \mapsto (\beta_{i_t}, \beta_{t_i})$; recover the one-axis Lorentz transform from a specified $\mathbb{R}^7$ lift-rotation-squeeze without fitting its coefficients after the fact; show how three paired slices reclose to the global Lorentz factor; determine whether d_i is merely notation or a literal differential light-distance/speed quantity; and recover the meaning of the original mixed angle φ_i without adding an independent degree of freedom to the one-axis state. Non-collinear boosts and Thomas-Wigner rotation are the obvious later stress test.

None of these obligations weakens the canon. They are what the canon now forces us to calculate.

Reference

National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, §4.23(viii), "Gudermannian Function," especially Eqs. 4.23.39–4.23.42.

Same relation. Different body. Same pair. Different projection.